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GTIIT

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LECTURE 18 - DECEMBER 11TH



MORE TECHNIQUES OF INTEGRATION

Rules of Integration of the Main Functions

$$F(x) = \int f(x)dx + C : C \in \mathbb{R}$$

$$1) \int 0 dx = C, \quad \int 1 dx = x$$

$$2) \int x^r dx = \frac{1}{r+1}x^{r+1}, r \neq -1, \text{ only for } x \in \text{Dom}(f), x \neq 0$$

$$3) \int \frac{1}{x} dx = \ln |x|, x \neq 0$$

$$4) \int e^x dx = e^x, \quad \int a^x dx = \frac{1}{\ln(a)}a^x$$

$$5) \int \sin x dx = -\cos x, \quad \int \cos x dx = \sin x$$

$$6) \int \frac{1}{\cos^2 x} dx = \tan x, \quad \int \frac{1}{\sin^2 x} dx = -\cot x$$

$$7) \int \cosh(x) dx = \sinh(x), \quad \int \sinh(x) dx = \cosh(x)$$

$$8) \int \frac{1}{a+x^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right)$$

$$9) \int \frac{1}{\sqrt{a^2-x^2}} dx = \arcsin\left(\frac{x}{a}\right), \quad \int \frac{-1}{\sqrt{a^2-x^2}} dx = \arccos\left(\frac{x}{a}\right)$$

$$10) \int \sec(x) \tan(x) dx = \sec(x), \quad \int \csc(x) \cot(x) dx = -\csc(x)$$

$$11) \int \sec(x) dx = \ln |\sec(x) + \tan(x)|, \quad \int \csc(x) dx = \ln |\csc(x) - \cot(x)|$$

$$12) \int \frac{1}{\sqrt{x^2 \pm a^2}} dx = \ln \left| x + \sqrt{x^2 \pm a^2} \right|$$

INTEGRATION BY PARTS

$$\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx (*)$$

Write $u = f(x)$ and $v = g(x)$, then the differentials are

$$du = f'(x)dx \text{ and } dv = g'(x)dx$$

Replace into $(*)$ to get

$$\int u dv = uv - \int v du$$

INTEGRATION BY PARTS

Solve $\int \sin(x) \cos(x) \ln(\sin(x)) dx$

Solution. Substitute $u = \sin(x) \implies du = \cos(x) dx$, then

$$\int \sin(x) \cos(x) \ln(\sin(x)) dx = \int u \cdot \ln(u) du$$

Integration by Parts:

$$\bullet u \, du = ds \text{ and } \ln(u) = t \implies \bullet \frac{1}{2}u^2 = s \text{ and } \frac{1}{u} du = dt$$

$$\implies \int \sin(x) \cos(x) \ln(\sin(x)) dx = \int u \cdot \ln(u) du$$

$$\Rightarrow \int \sin(x) \cos(x) \ln(\sin(x)) dx = \int u \cdot \ln(u) du$$

$$\boxed{\int t ds = s \cdot t - \int s dt} = \frac{1}{2} u^2 \ln(u) - \frac{1}{2} \int u^2 \frac{1}{u} du$$

$$= \frac{1}{2} u^2 \ln(u) - \frac{1}{4} u^2 + C$$

$$= \boxed{\frac{1}{2} \sin^2(x) \ln(\sin(x)) - \frac{1}{4} \sin^2(x) + C \checkmark}$$

Partial Fractions

- The strategy to integrate a rational function (a quotient of polynomials) p/q consists of expressing it as a sum of simpler fractions, called **partial fractions**, that we already know how to integrate.

- The degree of $p = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 + a_0$, is $\deg(p) = n$ if $a_n \neq 0$.

Condition for $\int \frac{p(x)}{q(x)} dx$:

- The degree of the numerator must be **less than** the degree of the denominator, otherwise, divide p by q
- The decomposition of p/q depends on the **factorization** of q .



Partial Fractions Case 1: Simple Roots

Ex 1. Solve $\int \frac{x^3 - 5x + 1}{(x^2 - 5)} dx$

Solution. • $\deg(p) = 3 \not< 2 = \deg(q)$, then divide p by q :

$$\begin{aligned}\frac{x^3 - 5x + 1}{(x^2 - 5)} &= \frac{x(x^2 - 5) + 1}{(x^2 - 5)} = \frac{x(x^2 - 5)}{(x^2 - 5)} + \frac{1}{(x^2 - 5)} \\ &= x + \frac{1}{(x^2 - 5)} = x + \frac{p}{q}\end{aligned}$$

- Factorize denominator by its **real roots**:

$$q = x^2 - 5 = (x - \sqrt{5})(x + \sqrt{5})$$

- Express p/q as a sum of **partial fractions**:

$$\frac{p}{q} = \frac{1}{(x^2 - 5)} = \frac{A}{(x - \sqrt{5})} + \frac{B}{(x + \sqrt{5})} = \frac{A(x + \sqrt{5}) + B(x - \sqrt{5})}{(x^2 - 5)}$$



Partial Fractions Case 1: Simple Roots

- Compare the numerators: the two polynomial coefficients must be equal:

$$1 = A(x + \sqrt{5}) + B(x - \sqrt{5}) = (A + B)x + \sqrt{5}(A - B)$$

$$\Rightarrow (A + B)x = 0 \quad \forall x \text{ and } \sqrt{5}(A - B) = 1$$

- Solve the linear system:

$$\begin{cases} A + B = 0 \\ \sqrt{5}(A - B) = 1 \end{cases} \Rightarrow B = -A \Rightarrow 2\sqrt{5}A = 1 \Rightarrow A = -B = \frac{1}{2\sqrt{5}}$$

- Express $x + p/q$ as a sum of partial fractions:

$$x + \frac{p}{q}(x) = x + \frac{1}{(x^2 - 5)} = \frac{1}{2\sqrt{5}} \frac{1}{(x - \sqrt{5})} - \frac{1}{2\sqrt{5}} \frac{1}{(x + \sqrt{5})}$$

Partial Fractions Case 1: Simple Roots

- Integrate the integrand expressed in **partial fractions**:

$$\begin{aligned}\int \frac{x^3 - 5x + 1}{(x^2 - 5)} dx &= \int \left[x + \frac{1}{(x^2 - 5)} \right] dx \\&= \int x \, dx + \frac{1}{2\sqrt{5}} \int \frac{dx}{(x - \sqrt{5})} - \frac{1}{2\sqrt{5}} \int \frac{dx}{(x + \sqrt{5})} \\&= \boxed{\frac{1}{2}x^2 + \frac{1}{2\sqrt{5}} \ln |x - \sqrt{5}| - \frac{1}{2\sqrt{5}} \ln |x + \sqrt{5}| + C} \checkmark\end{aligned}$$

Theorem: Solve Partial Fractions Case 1, Simple Roots

Given a quotient $f(x) = \frac{p(x)}{q(x)}$ with $\deg(p) < \deg(q)$ such that q factorizes in distinct linear factors

$$q(x) = (a_1x + b_1) \dots (a_nx + b_n)$$

where no factor is a constant multiple of another, then there exist constants $A_1, A_2, \dots, A_n \in \mathbb{R}$ such that

$$f(x) = \frac{p(x)}{q(x)} = \frac{A_1}{(a_1x + b_1)} + \frac{A_2}{(a_2x + b_2)} + \dots + \frac{A_n}{(a_nx + b_n)}$$

Theorem: Solve Partial Fractions Case 2, Repeated Linear Factors

Given a quotient of polynomial $\frac{p(x)}{q(x)}$ with $\deg(p) < \deg(q)$ such that q factorizes in the form

$$q(x) = (a_1x + b_1) \cdots (a_{n-1}x + b_{n-1})(a_nx + b_n)^r$$

then there exist constants $A_1, A_2, \dots, A_{n-1}, B_1, \dots, B_r \in \mathbb{R}$ such that

$$\begin{aligned} \frac{p(x)}{q(x)} = & \frac{A_1}{(a_1x + b_1)} + \cdots + \frac{A_{n-1}}{(a_{n-1}x + b_{n-1})} \\ & + \frac{B_1}{(a_nx + b_n)} + \frac{B_2}{(a_nx + b_n)^2} + \cdots + \frac{B_r}{(a_nx + b_n)^r} \end{aligned}$$

Analogously, add ℓ terms with increasing exponents, for each factor of the form $(ax + b)^\ell$



Partial Fractions Case 2: Repetead Linear Factors

Ex 2. Solve $\int \frac{4x}{(x^3 - x^2 - x + 1)(x - 1)} dx$

Solution. • $\deg(p) = 1 < 4 = \deg(q)$ ✓

- Factorize the denominator by its **real roots**:

$$q = (x^3 - x^2 - x + 1)(x - 1) = (x - 1)^2(x + 1)(x - 1) \\ = (x - 1)^3(x + 1)$$

- Express p/q as a sum of **partial fractions**:

$$\frac{p}{q} = \frac{4x}{(x^3 - x^2 - x + 1)} = \frac{A}{(x + 1)} + \frac{B}{(x - 1)} + \frac{C}{(x - 1)^2} + \frac{D}{(x - 1)^3}$$

$$= \frac{A(x - 1)^3 + B(x - 1)^2(x + 1) + C(x - 1)(x + 1) + D(x + 1)}{(x + 1)(x - 1)^3}$$

Partial Fractions Case 2

- Compare the numerators: the two **polynomial coefficients** must be equal:

$$\begin{aligned} 4x &= A(x-1)^3 + B(x-1)^2(x+1) + C(x-1)(x+1) + D(x+1) \\ &= (A+B)x^3 + (-3A-B+C)x^2 + (3A-B+D)x - A + B - C + D \end{aligned}$$

- Solve the linear system:

$$\begin{cases} A+B = 0 \implies B = -A \\ -3A-B+C = 0 \implies C = 2A \\ 3A-B+D = 4 \implies D = 4-4A \\ -A+B-C+D = 0 \implies D = 2A+C \implies A = -B = \frac{1}{2}, C = 1, D = 2 \end{cases}$$

- Express $f(x) = p/q$ as a sum of **partial fractions**:

$$\frac{4x}{(x^3 - x^2 - x + 1)} = \frac{1}{2(x+1)} - \frac{1}{2(x-1)} + \frac{1}{(x-1)^2} + \frac{2}{(x-1)^3}$$

Solve the integral using the decomposition:

- Integrate the function expressed in **partial fractions**:

$$\int \frac{4x}{(x^3 - x^2 - x + 1)}$$

$$= \int \frac{1}{2(x+1)} dx - \int \frac{1}{2(x-1)} dx + \int \frac{1}{(x-1)^2} dx + \frac{2}{(x-1)^3} dx$$

$$= \frac{1}{2} \ln |x+1| - \frac{1}{2} \ln |x-1| - \frac{1}{(x-1)} dx - \frac{1}{(x-1)^2} \checkmark$$

Theorem: Solve Partial Fractions Case 3, where q contains Quadratic Factors, none of which is repeated

Under the conditions of Case 2, if q has a factor $ax^2 + bx + c$ with no real roots, that is, where $b^2 - 4ac < 0$, then, in addition to the partial fractions of Case 2, the expression for p/q will have a term of the form

$$\frac{Ax + B}{ax^2 + bx + c}$$

where A and B are constants to be determined.
Add one term of this form for each different quadratic factor of q with no real roots.

$$\text{Ex. } \frac{x}{(x-2)(x^2+1)(x^2+3)} = \frac{A}{(x-2)} + \frac{Bx+C}{(x^2+1)} + \frac{Dx+E}{(x^2+4)}$$

Partial Fractions Case 3.

Ex 4. Solve $\int \frac{2x^2 - x + 4}{(x^3 + 4x)} dx$

Solution. • $\deg(p) = 2 < 3 = \deg(q)$ ✓

- Factorize the denominator by its **real roots**:

$$q = x^3 + 4x = x(x^2 + 4)$$

- Express p/q as a sum of **partial fractions**:

$$\frac{2x^2 - x + 4}{(x^3 + 4x)} = \frac{A}{x} + \frac{(Bx + C)}{(x^2 + 4)}$$

$$= \frac{A(x^2 + 4) + (Bx + C)x}{x(x^2 + 4)} = \frac{(A + B)x^2 + Cx + 4A}{x(x^2 + 4)}$$

Partial Fractions Case 3: Quadratic Factors

- Compare the numerators: the **three polynomial coefficients** must be equal:

$$2x^2 - x + 4 = (A + B)x^2 + Cx + 4A$$

- Solve the linear system:

$$\begin{cases} A + B = 0 & \implies B = -A \\ 2 = A + B & \implies B = 2 - A \\ -1 = C \\ 4 = 4A & \implies 1 = A \implies B = 1, C = -1 \end{cases}$$

Then, the integrand is written in **partial fractions**:

$$\frac{2x^2 - x + 4}{(x^3 + 4x)} = \frac{1}{x} + \frac{(x - 1)}{(x^2 + 4)}$$

Partial Fractions Case 3: Quadratic Factors

So, the integral can be solved in the way:

$$\int \frac{2x^2 - x + 4}{(x^3 + 4x)} dx = \int \frac{1}{x} dx + \int \frac{(x - 1)}{(x^2 + 4)} dx$$

$$\begin{aligned} &= \int \frac{1}{x} dx + \int \frac{x}{(x^2 + 4)} dx - \int \frac{1}{(x^2 + 4)} dx \\ &= \int \frac{1}{x} dx + \frac{1}{2} \int \frac{2x}{(x^2 + 4)} dx - \frac{1}{4} \int \frac{1}{\left(\left(\frac{x}{2}\right)^2 + 1\right)} dx \end{aligned}$$

$$= \ln |x| + \frac{1}{2} \ln |x^2 + 4| - \frac{1}{2} \arctan \left(\frac{x}{2} \right) + C \quad \checkmark$$

Case 3: Complete Squares in the Denominator

Ex 5. Solve $\int \frac{4x^2 - 3x + 2}{(4x^2 - 4x + 3)} dx$

Solution. • $\deg(p) = 2 \not< 2 = \deg(q)$, so, divide p by q

$$\begin{aligned}\frac{p}{q} &= \frac{4x^2 - 3x - x + 1 + x - 1 + 2}{(4x^2 - 4x + 3)} = \frac{(4x^2 - 4x + 3)}{(4x^2 - 4x + 3)} + \frac{x - 1}{(4x^2 - 4x + 3)} \\ &= 1 + \frac{x - 1}{(4x^2 - 4x + 3)}\end{aligned}$$

- The denominator has **no real roots**:

$$b^2 - 4a = 16 - 16 \times 3 < 0 \implies \text{Complete Squares}$$

$$q = (4x^2 - 4x + 3) = (2x - 1)^2 + 2$$

- If $u = (2x - 1) \implies (2x - 1)^2 + 2 = u^2 + 2$ and $du = 2dx$

In this case there is no further decomposition, since the denominator is already a simple fraction

Substitute in the given integral:

$$\begin{aligned}\int \frac{4x^2 - 3x + 2}{(4x^2 - 4x + 3)} dx &= \int 1 dx + \int \frac{x - 1}{(4x^2 - 4x + 3)} dx \\&= x + \int \frac{\frac{1}{2}(u + 1) - 1}{(u^2 + 2)} \frac{1}{2} du \\&= x + \frac{1}{4} \int \frac{(u - 1)}{(u^2 + 2)} du = x + \frac{1}{4} \int \frac{u}{(u^2 + 2)} du - \frac{1}{4} \int \frac{1}{(u^2 + 2)} du \\&= x + \frac{1}{8} \int \frac{2u}{(u^2 + 2)} du - \frac{1}{8} \int \frac{1}{\left(\left(\frac{u}{\sqrt{2}}\right)^2 + 1\right)} du\end{aligned}$$

Solve the Integral

$$= x + \frac{1}{8} \int \frac{2u}{(u^2 + 2)} du - \frac{1}{8} \int \frac{1}{\left(\left(\frac{u}{\sqrt{2}}\right)^2 + 1\right)} du$$

$$= x + \frac{1}{8} \ln |u^2 + 2| - \frac{1}{8} \int \frac{1}{\left(\left(\frac{u}{\sqrt{2}}\right)^2 + 1\right)} du$$

$$= x + \frac{1}{8} \ln |u^2 + 2| - \frac{\sqrt{2}}{8} \arctan \left(\frac{u}{\sqrt{2}} \right) + C$$

Substitute back $u = (2x - 1)$:

$$= x + \frac{1}{8} \ln |(2x - 1)^2 + 2| - \frac{\sqrt{2}}{8} \arctan \left(\frac{(2x-1)}{\sqrt{2}} \right) + C \checkmark$$

Remark on Case 3

In general, to solve $\int \frac{Ax + B}{(ax^2 + bx + c)} dx$, where $b^2 - 4ac < 0$:

- Complete squares in the denominator:

$$(ax^2 + bx + c) = (u^2 + d^2)$$

- After rewriting, solve the integral in the way:

$$\begin{aligned} \int \frac{Ax + B}{(ax^2 + bx + c)} dx &= \int \frac{Cu + D}{(u^2 + d^2)} dx \\ &= \frac{C}{2} \int \frac{2u}{(u^2 + d^2)} dx + \frac{D}{d^2} \int \frac{1}{((u/d)^2 + 1)} dx \\ &= \frac{C}{2} \ln(u^2 + d^2) + \frac{D}{d^2} \arctan(u/d) \quad \checkmark \end{aligned}$$

Theorem: Solve Partial Fractions in Case 4, where q contains a repeated irreducible (Quadratic) Factor

Under the conditions of Case 2, if q has a factor $(ax^2 + bx + c)^r$ with no real roots, that is, where $b^2 - 4ac < 0$, then, in addition to the partial fractions of Case 2, the expression for p/q has terms the form

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_rx + B_r}{(ax^2 + bx + c)^r} +$$

where A_k and $B_k : 1 \leq k \leq r$ are constants to be determined.

Ex 6. Express the form of the partial fractions of

$\frac{p}{x(x-2)(x^2+x+1)(x^2+3)^3}$, where p is a polynomial of degree < 10 .

Solution. The factors (x^2+x+1) and $(x^2+3)^3$ have no real roots, then the decomposition takes the following form:

$$\frac{p}{x(x-2)(x^2+x+1)(x^2+3)^3} = \frac{A}{x} + \frac{B}{(x-2)} + \frac{Cx+D}{(x^2+x+1)} + \frac{Ex+F}{(x^2+3)} + \frac{Gx+H}{(x^2+3)^2} + \frac{Ix+J}{(x^2+3)^3} \quad \checkmark$$

Ex 7. Solve by means of rationalizing substitutions

$$\int \frac{\sqrt{x+9}}{x} dx$$

- Some **nonrational functions** can be changed into **rational functions** by means of appropriate substitutions.
- If an integrand contains an expression of the form $\sqrt[n]{g(x)}$, then the substitution $u = \sqrt[n]{g(x)}$ may be effective.

Solution. • Let $u = \sqrt{x+9} \implies u^2 = x+9 \implies x = u^2 - 9$

- $dx = 2u \, du$, then

$$\int \frac{\sqrt{x+9}}{x} dx = \int \frac{2u^2}{u^2-9} du = 2 \int \left[1 + \frac{9}{u^2-9} \right] du$$

Partial Fractions

- We need a partial fraction decomposition for the last:

$$\frac{9}{u^2 - 9} = \frac{A}{u - 3} + \frac{B}{u + 3} = \frac{A(u + 3) + B(u - 3)}{u^2 - 9}$$

- The numerators are the same:

$$\implies 9 = A(u + 3) + B(u - 3)$$

- To find the coefficients, since the identity is true for **all** $x \in \mathbb{R}$, evaluate on convenient values, namely, **on the roots of the denominator**:

$$\bullet 9 = 6A \implies A = \frac{3}{2}$$

$$\bullet 9 = -6B \implies B = -\frac{3}{2}$$

Solve the Integral

Then, the integrand expresses in the form

$$\int \frac{\sqrt{x+9}}{x} dx = 2 \int \left[1 + \frac{9}{u^2 - 9} \right] du$$

$$= 2u + 2 \int \frac{3}{2(u-3)} du - 2 \int \frac{3}{2(u+3)} du$$

$$= 2u + 3 \ln |u - 3| - \ln |u + 3| + C$$

$$= 2\sqrt{x+9} + 3 \ln |\sqrt{x+9} - 3| - \ln |\sqrt{x+9} + 3| + C \checkmark$$

IMPROPER INTEGRALS

Type 1: Infinite Intervals

Consider the infinite region R that lies under the curve $y = \frac{1}{x^2}$, above the x -axis, and to the right of the line $x = 1$. Is this area finite or infinite? On what does it depend?

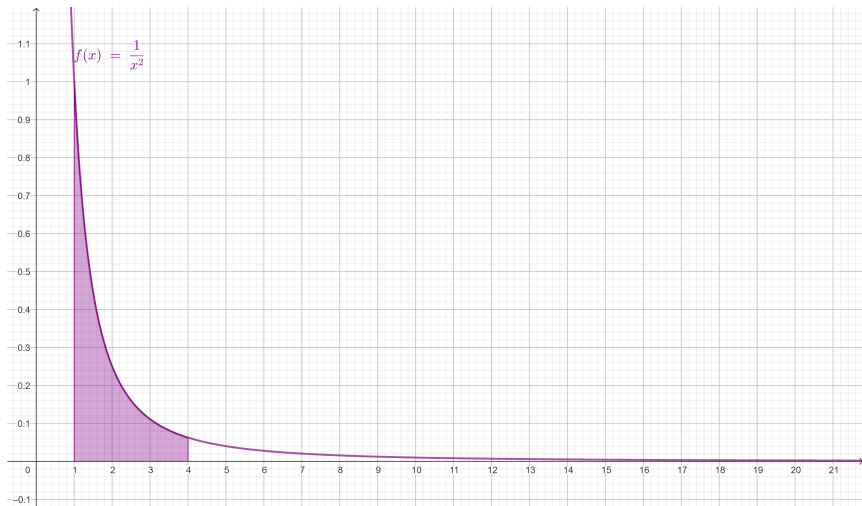
- The area of the region R that lies under the curve $y = \frac{1}{x^2}$, above the x -axis, to the right of the line $x = 1$ and to the left of the line $x = t$, is:

$$\text{Area } A(t) = \int_1^t \frac{1}{x^2} dx = -\frac{1}{x} \Big|_1^t = 1 - \frac{1}{t}$$

- Notice that, no matter how large t is chosen,

$$\text{Area on } [1, t]: 0 < A(t) < 1$$

Improper Integral of Type 1: Infinite Intervals



Area below the curve on $[1, t]$:
$$= \int_1^t \frac{1}{x^2} dx$$

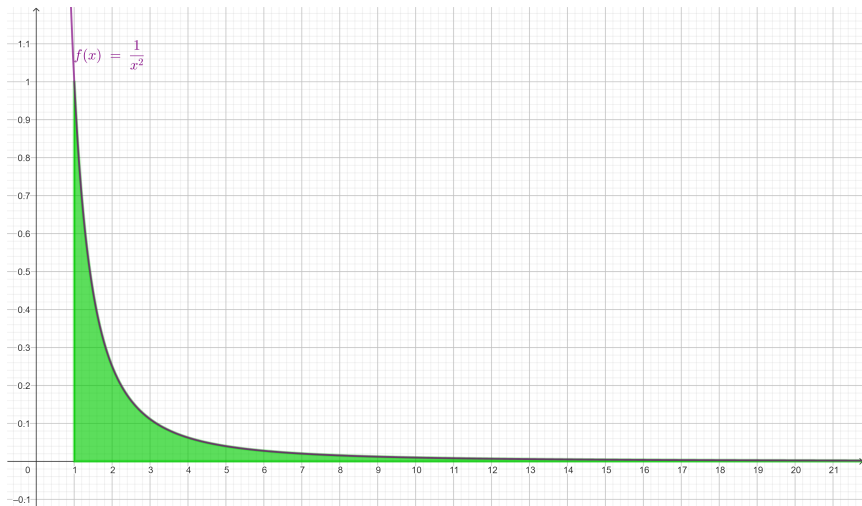
Type 1: Infinite Intervals

- The area of the shaded region approaches 1 as $t \rightarrow \infty$, so we say that the area of the infinite region R is equal to 1 and we write:

$$\text{Area below the curve on } [1, +\infty) = \int_1^{\infty} \frac{1}{x^2} dx$$

$$= \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \left[-\frac{1}{x} \Big|_1^t \right] = \lim_{t \rightarrow \infty} \left[1 - \frac{1}{t} \right] = 1$$

Improper Integral of Type 1: Infinite Intervals



Improper Integral of $f(x) = \frac{1}{x^2}$ on $[1, \infty)$: $\int_1^{\infty} \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2} dx = 1$

Definition of Improper Integral of Type 1: Infinite Intervals

- 1 If $\int_a^t f(x)dx$ exists for every $t \geq a$, then we define

$$\int_a^\infty f(x)dx = \lim_{t \rightarrow \infty} \int_a^t f(x)dx$$

provided this limit exists (as a finite number).

- 2 If $\int_t^b f(x)dx$ exists for every $t \leq b$, then we define

$$\int_{-\infty}^b f(x)dx = \lim_{t \rightarrow -\infty} \int_t^b f(x)dx$$

provided this limit exists (as a finite number).

In any of these cases, the improper integrals are called **convergent**. It is called **divergent** if the limit doesn't exist in any sense, in particular, if it is $\pm\infty$.



Definition of Improper Integral of Type 1: Infinite Interval $(-\infty, +\infty)$

If for any $a \in \mathbb{R}$, both improper integrals

$$\int_{-\infty}^a f(x)dx \quad \text{and} \quad \int_a^{\infty} f(x)dx$$

are **convergent**, then we define

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^a f(x)dx + \int_a^{\infty} f(x)dx$$

Notice that, by definition, both limits must exist, independently, in order to get the convergence of

$$\int_{-\infty}^{\infty} f(x)dx$$

Improper Integral of Type 1: Infinite Intervals

Ex 8. Determine whether the improper integral is **convergent** or **divergent**:

$$\int_1^{\infty} \frac{1}{x} dx$$

Solution.

$$\int_1^{\infty} \frac{1}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} \left[\ln |x| \Big|_1^t \right]$$

$$= \lim_{t \rightarrow \infty} [\ln(t) - \ln(1)] = \lim_{t \rightarrow \infty} \ln(t) = \infty$$

Then, the improper integral is **divergent**

Improper Integral of Type 1: Infinite Intervals

Let's compare the result of the two examples:

$$\int_1^{\infty} \frac{1}{x^2} dx \text{ converges and } \int_1^{\infty} \frac{1}{x} dx = \infty \text{ diverges}$$

- Geometrically, this says that although the curves $y = \frac{1}{x}$ and $y = \frac{1}{x^2}$ look very similar, the region under $y = \frac{1}{x^2}$ to the right of $x = 1$ has **finite area** whereas the corresponding region under $y = \frac{1}{x}$ has **infinite area**.
- Note that both $y = \frac{1}{x^2}$ and $y = \frac{1}{x}$ approach 0 as $x \rightarrow \infty$ but $y = \frac{1}{x^2}$ approaches 0 faster than $y = \frac{1}{x}$. The values of $y = \frac{1}{x}$ **don't decrease fast enough** for its integral to have a finite value.

Improper Integral of Type 1: Infinite Intervals

Ex 9. Evaluate the improper integral

$$\int_{-\infty}^0 x e^x dx$$

Solution. By definition:

$$\int_{-\infty}^0 x e^x dx = \lim_{t \rightarrow -\infty} \int_t^0 x e^x dx$$

Integrate by parts in order to solve the antiderivative, let:

- $u = x$, $dv = e^x dx$, then $du = dx$, $v = e^x$, so

$$\int_t^0 x e^x dx = (x e^x) \Big|_t^0 - \int_t^0 e^x dx = -t e^t - 1 + e^t$$

Improper Integral of Type 1: Infinite Intervals

Let us evaluate the limit by **L'Hospital Rule** and notice

$$\lim_{t \rightarrow +\infty} e^t = +\infty \implies \lim_{t \rightarrow -\infty} e^{-t} = +\infty$$

$$\implies \lim_{t \rightarrow +\infty} t e^t = \lim_{t \rightarrow -\infty} \frac{t}{e^{-t}} = \lim_{t \rightarrow -\infty} \frac{1}{-e^{-t}} = 0$$

Hence, for the improper integral, we get:

$$\int_{-\infty}^0 x e^x dx = \lim_{t \rightarrow -\infty} \int_t^0 x e^x dx$$

$$\lim_{t \rightarrow -\infty} \left[-t e^t - 1 + e^t \right] = 0 - 1 + 0 = -1 \checkmark$$

So, the improper integral is convergent



Improper Integral of Type 1: Infinite Intervals

Ex 10. Evaluate the improper integral

$$\int_{-\infty}^{+\infty} \frac{1}{1+x^2} dx$$

Solution. By definition, we can choose any $c \in \mathbb{R}$ and let $c = 0$:

$$\int_{-\infty}^{+\infty} \frac{1}{1+x^2} dx = \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{+\infty} \frac{1}{1+x^2} dx$$

$$= \lim_{t \rightarrow -\infty} \int_t^0 \frac{1}{1+x^2} dx + \lim_{t \rightarrow +\infty} \int_0^t \frac{1}{1+x^2} dx$$

Integrate in order to solve the antiderivative:

$$\int \frac{1}{1+x^2} dx = \arctan(x)$$

Improper Integral of Type 1: Infinite Intervals

Evaluate then the limit of each integral separately:

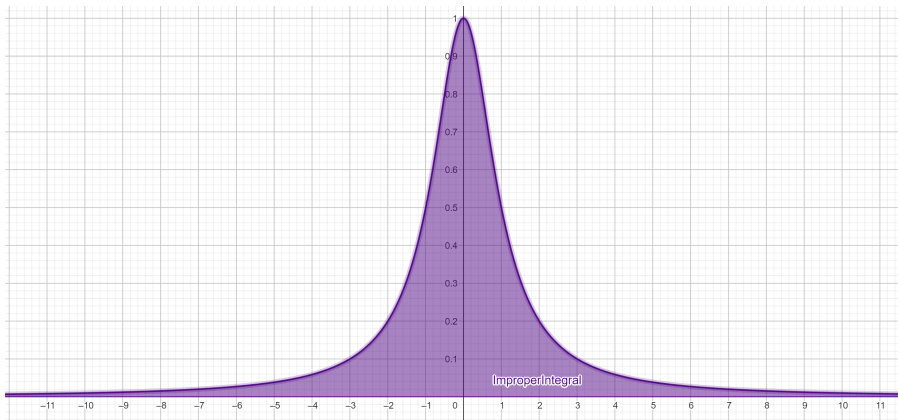
$$\int_{-\infty}^0 \frac{1}{1+x^2} = \lim_{t \rightarrow -\infty} \arctan(x) \Big|_t^0 = \lim_{t \rightarrow -\infty} [\arctan(0) - \arctan(t)] = - \left[-\frac{\pi}{2} \right]$$

$$\int_0^{+\infty} \frac{1}{1+x^2} = \lim_{t \rightarrow +\infty} \arctan(x) \Big|_0^t = \lim_{t \rightarrow +\infty} [\arctan(t) - \arctan(0)] = \frac{\pi}{2}$$

So, the improper integral is convergent:

$$\int_{-\infty}^{+\infty} \frac{1}{1+x^2} dx = \int_{-\infty}^0 \frac{1}{1+x^2} + \int_0^{+\infty} \frac{1}{1+x^2} = \pi \checkmark$$

Improper Integral of Type 1: Infinite Intervals



Improper Integral of $f(x) = \frac{1}{1+x^2}$ on $(-\infty, \infty)$:

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx = \pi \quad \checkmark$$

Proposition: Integrals of type p

The improper integral

$$\int_1^{+\infty} \frac{1}{x^p} dx \text{ converges} \iff p > 1$$

In this case,

$$\int_1^{+\infty} \frac{1}{x^p} dx = \frac{1}{(p-1)}$$

This means that, for all $p > 1$, the integral converges and it diverges for all $p \leq 1$.

Proof: Case $p \neq 1$

Case $p \neq 1$. By definition

$$\int \frac{1}{x^p} dx = \int x^{-p} dx = \frac{1}{(-p+1)} x^{-p+1} = \frac{1}{(-p+1)} \frac{1}{x^{p-1}}$$

$$\Rightarrow \int_1^{+\infty} \frac{1}{x^p} dx = \lim_{t \rightarrow +\infty} \int_1^t \frac{1}{x^p} dx = \lim_{t \rightarrow +\infty} \left[\frac{1}{(-p+1)} \frac{1}{x^{p-1}} \right] \Big|_{x=1}^{x=t}$$

• **Case $p > 1$** , we have $\lim_{t \rightarrow +\infty} \frac{1}{t^{p-1}} = 0$

$$\Rightarrow \int_1^{+\infty} \frac{1}{x^p} dx = \frac{1}{(-p+1)} \lim_{t \rightarrow +\infty} \left[\frac{1}{t^{p-1}} - 1 \right] = \frac{1}{(p-1)}$$

Proposition: Integrals of type p for $p < 1$

- **Case $p < 1$:** in this case, the previous limit is $+\infty$:

$$\lim_{t \rightarrow +\infty} \left[\frac{1}{(-p+1)} \frac{1}{x^{p-1}} \right] \Big|_{x=1}^{x=t} = \lim_{t \rightarrow +\infty} \left[\frac{1}{(1-p)} \left(t^{1-p} \right)^{\rightarrow \infty} - \frac{1}{(1-p)} \right] = \infty$$

So, the **improper integral diverges for $p < 1$.**

Proposition: Integrals of type p for $p = 1$

- **Case $p = 1$:** The antiderivative is not the same:

$$\int \frac{1}{x^1} dx = \ln |x|$$
$$\Rightarrow \int_1^{+\infty} \frac{1}{x} dx = \lim_{t \rightarrow +\infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow +\infty} \left[\ln |x| \right]_{x=1}^{x=t}$$
$$= \lim_{t \rightarrow +\infty} [\ln t] - \ln(1) = +\infty$$

so, the **improper integral diverges for $p = 1$** ✓

Hence, the improper integral converges only for $p > 1$, and, in this case,

$$\int_1^{+\infty} \frac{1}{x^p} dx = \lim_{t \rightarrow +\infty} \left[\frac{1}{(1-p)} \frac{1}{x^{p-1}} \right] \Big|_{x=1}^{x=t} = \frac{1}{(1-p)} \quad \checkmark$$

Example 2. a)

For $p = 5$ and $p = \frac{3}{2}$ the following integrals converge:

$$i) \int_1^{+\infty} \frac{1}{x^5} dx = \frac{1}{4};$$

$$ii) \int_1^{+\infty} \frac{1}{x^{\frac{3}{2}}} dx = \frac{1}{(\frac{3}{2} - 1)} = 2$$

Example 2. b)

For $p = \frac{1}{2} < 1$, the integral diverges:

$$\begin{aligned}\int_1^{+\infty} \frac{1}{\sqrt{x}} dx &= \lim_{t \rightarrow +\infty} \int_1^t \frac{1}{x^{\frac{1}{2}}} dx = \lim_{t \rightarrow +\infty} [2\sqrt{x}] \Big|_{x=1}^{x=t} \\ &= \lim_{t \rightarrow +\infty} [2(\sqrt{t} - 1)] = +\infty\end{aligned}$$

Improper Integral of Type 2: Discontinuous Integrand

- Suppose that f is a continuous function defined on a **bounded interval** $[a, b)$ but f has a vertical asymptote at b .
- If f is positive, let R be the **unbounded** region **under the graph of f and above the x -axis between a and b** in a horizontal direction. Here the region is extended indefinitely in a vertical direction.
- The area of the part of R between a and t is

$$A(t) = \int_a^t f(x) dx$$

- If $A(t) \rightarrow L \in \mathbb{R}$, that is, it approaches a finite number as $t \rightarrow b^-$, then we say that the improper integral converges.
- The same definition holds for other types of discontinuities.

Improper Integral of Type 2: Discontinuous Integrand

- 1 If f is a continuous function defined on a **bounded interval** $[a, b)$ and discontinuous at b , we define

$$\int_a^b f(x)dx = \lim_{t \rightarrow b^-} \int_a^t f(x)dx$$

provided this limit exists (as a finite number).

- 2 If f is a continuous function defined on a **bounded interval** $(a, b]$ and discontinuous at a , we define

$$\int_a^b f(x)dx = \lim_{t \rightarrow a^+} \int_t^b f(x)dx$$

provided this limit exists (as a finite number).

In these cases, the integral is called **convergent**.

It is called **divergent** if the limit doesn't exist in any sense.



Improper Integral of Type 2: Discontinuous Integrand

If f is a continuous function defined on $[a, c)$ and on $(c, b]$, that is, if f is discontinuous at c , and both, $\int_a^c f(x)dx$ and $\int_c^b f(x)dx$ are convergent, we define

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

In the case of a convergent integral, the region is unbounded but its area is finite.

Improper Integral of Type 2: Discontinuous Integrand

Ex. 11. Compute $\int_3^7 \frac{1}{\sqrt{x-3}} dx$

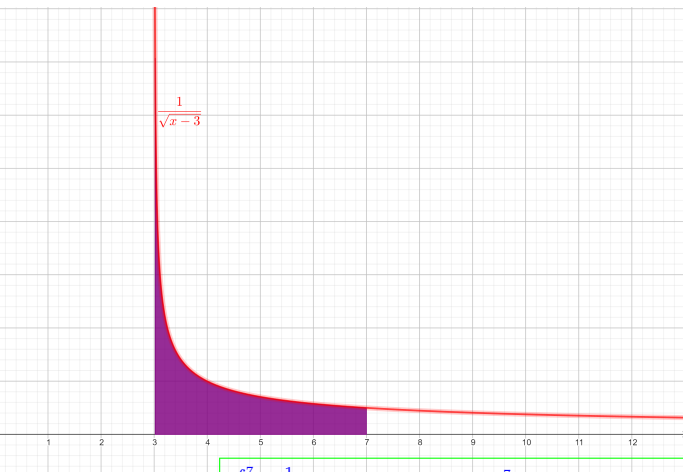
- Solution. We note first that the given integral is **improper** because $f(x) = \frac{1}{\sqrt{x-3}}$ has the vertical asymptote $x = 3$.
- Since the infinite discontinuity occurs at the left:

$$\int_3^7 \frac{1}{\sqrt{x-3}} dx = \lim_{t \rightarrow 3^+} \int_t^7 \frac{1}{\sqrt{x-3}} dx$$

$$= \lim_{t \rightarrow 3^+} \left[2\sqrt{x-3} \right] \Big|_t^7 = \lim_{t \rightarrow 3^+} 2 \left[\sqrt{4} - \sqrt{t-3} \right] = 4 \checkmark$$

So, the integral **converges** $\checkmark$

Improper Integral of Type 2: Discontinuous Integrand



$$\int_3^7 \frac{1}{\sqrt{x-3}} dx = \lim_{t \rightarrow 3^+} \left[2\sqrt{x-3} \right]_t^7 = \lim_{t \rightarrow 3^+} \left[4 - 2\sqrt{t-3} \right] = 4 \checkmark$$

Improper Integral of Type 2: Discontinuous Integrand

Ex. 12. Decide whether the improper integral

$\int_0^{\pi/2} \sec(x) dx$ converges or diverges

- Solution. Note that the given integral is **improper** because $f(x) = \sec(x)$ has the vertical asymptote $x = \pi/2$:

$$\lim_{t \rightarrow \pi/2^-} \sec(x) = \infty$$

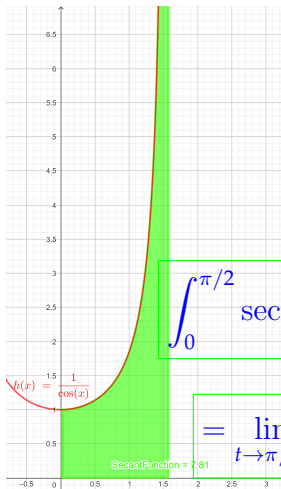
- Since the infinite discontinuity occurs at the right:

$$\int_0^{\pi/2} \sec(x) dx = \lim_{t \rightarrow \pi/2^-} \int_0^t \sec(x) dx = \lim_{t \rightarrow \pi/2^-} \ln |\sec(x) + \tan(x)| \Big|_0^t$$

$$= \lim_{t \rightarrow \pi/2^-} \ln |\sec(t) + \tan(t)| - \ln(1) = \lim_{t \rightarrow \pi/2^-} \ln \left| \frac{1+\sin t}{\cos t} \right|^{\rightarrow \infty} = \infty \quad \checkmark$$

So, the improper integral **diverges** ✓

Improper Integral of Type 2: Discontinuous Integrand



$$\int_0^{\pi/2} \sec(x) dx = \lim_{t \rightarrow \pi/2^-} = \lim_{t \rightarrow \pi/2^-} \ln |\sec(x) + \tan(x)| \Big|_0^t$$

$$= \lim_{t \rightarrow \pi/2^-} \left[\ln |\sec(t) + \tan(t)| - \ln(1) \right] = \infty$$

Improper Integral of Type 2: Discontinuous Integrand

Ex. 13. a) Compute, when possible, $\int_0^4 \frac{1}{x-1} dx$

- **Solution.** Is the following statement true or false?
Does the improper integral **converge**?

$$\int_0^4 \frac{1}{x-1} dx = [\ln |x-1|] \Big|_0^4 = [\ln |3| - \ln |1|] = \ln(3) \quad \checkmark$$

Attention: this is wrong, because the integrand has a vertical asymptote in the interval of integration, so, the integral is **improper**

$$\int_0^4 \frac{1}{x-1} dx = \cancel{\ln |x-1|} \Big|_0^4 = \cancel{\ln |3| - \ln |1|} = \ln(3) \quad \checkmark$$

Improper Integral of Type 2: Discontinuous Integrand

Ex. 13. b) Decide whether the improper integral

$$\int_0^4 \frac{1}{x-1} dx \text{ converges or diverges}$$

• **Right Solution.** Note that the given integral is **improper** because $f(x) = \frac{1}{x-1}$ has the vertical asymptote $x = 1$:

$$\lim_{t \rightarrow 1^\pm} \frac{1}{x-1} = \pm\infty$$

• The discontinuity occurs **in the middle of the interval**: f is **continuous** on $[0, 1)$ and on $(1, 4]$

Improper Integral of Type 2: Discontinuous Integrand

- Since the infinite discontinuity occurs **in the middle of the interval**, by definition:

$$\int_0^4 \frac{1}{x-1} dx = \int_0^1 \frac{1}{x-1} dx + \int_1^4 \frac{1}{x-1} dx$$

Consider separately the 2nd term, where $x > 1$:

$$\int_1^4 \frac{1}{x-1} dx = \lim_{t \rightarrow 1^+} \ln|x-1| \Big|_t^4 = \lim_{t \rightarrow 1^+} \left[\ln(3) - \ln|t-1|_{\rightarrow -\infty} \right] = \infty \quad \checkmark$$

So, the improper integral **diverges** $\checkmark$
(and there is no need to evaluate the 1st term)

Comparison Test for Improper Integrals

Suppose that f and g **positive, continuous** functions with $f(x) \geq g(x) \geq 0$ for all $x \geq a$:

- 1 If $\int_a^\infty f(x)dx$ converges, then $\int_a^\infty g(x)dx$ converges
- 2 If $\int_a^\infty g(x)dx$ diverges, then $\int_a^\infty f(x)dx$ diverges

Comparison Test for Improper Integrals

Ex. 14. Prove that $\int_1^{\infty} e^{-x^2} dx$ is convergent

Solution: $g(x) = e^{-x^2} \leq f(x) = e^{-x}$ and f, g are **positive, continuous** functions for all $x \geq 1$, then:

$$0 \leq \int_1^{\infty} e^{-x^2} dx \leq \int_1^{\infty} e^{-x} dx$$

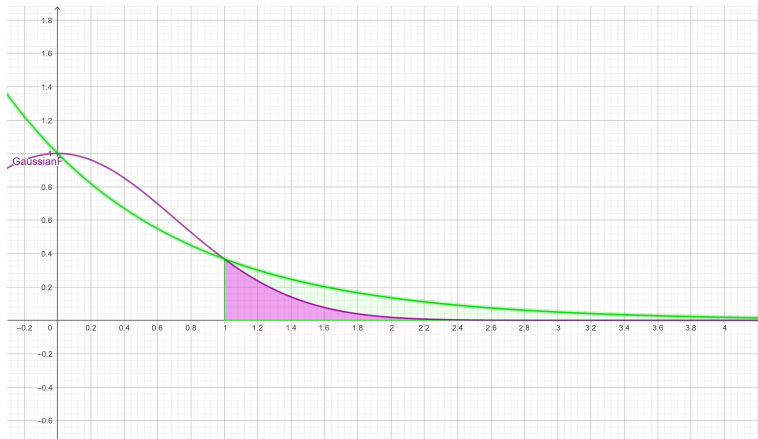
We can solve the integral on the right: there is an **elementary antiderivative**

$$\int_1^{\infty} e^{-x} dx = \lim_{t \rightarrow \infty} \int_1^t e^{-x} dx = \lim_{t \rightarrow \infty} \left[-e^{-x} \right]_1^t = \lim_{t \rightarrow \infty} \left[-e^{-t} + e^{-1} \right] = e^{-1}$$

$\Rightarrow$ the integral converges, since it is bounded by a convergent one

$$0 \leq \underbrace{\int_1^{\infty} e^{-x^2} dx}_{\text{convergent}} \leq \int_1^{\infty} e^{-x} dx = e^{-1}$$

Comparison Test for Improper Integrals



$$0 \leq \int_1^\infty e^{-x^2} dx \leq \int_1^\infty e^{-x} dx = e^{-1}$$

Comparison Test for Improper Integrals

Ex. 15. Prove that $\int_1^{\infty} \frac{1 + e^{-x}}{x} dx$ is divergent

Solution: • Compare f with a convenient g and use the Comparison Test:

$$e^{-x} > 0 \implies 1 + e^{-x} > 1$$

- Then, divide by the positive factor x :

$$\implies f(x) = \frac{1 + e^{-x}}{x} \geq g(x) = \frac{1}{x}$$

- f, g are **positive, continuous** functions for all $x \geq 1$, then:

$$\int_1^{\infty} \frac{1 + e^{-x}}{x} dx \geq \int_1^{\infty} \frac{1}{x} dx = \lim_{t \rightarrow \infty} (\ln |t| - \ln(1)) = +\infty$$

$\implies$ the largest integral must be ∞ as well, and so, it diverges ✓

Improper Integrals

Ex. 16. Decide whether the improper integral converges or diverges; if it converges, find its value: $\int_0^1 \ln(x) dx$

Solution: • The given integral is improper, since $\lim_{t \rightarrow 0^+} \ln(t) = -\infty$, so by definition

$$\int_0^1 \ln(x) dx = \lim_{t \rightarrow 0^+} \int_t^1 \ln(x) dx$$

- We integrate by parts with $u = \ln x$, $dv = dx$ and so, $du = \frac{1}{x} dx$, and $v = x$:

$$\int \ln(x) dx = x \ln x - \int 1 dx = x \ln x - x$$

Improper Integrals

- Evaluate the integral:

$$\int_0^1 \ln(x) dx = \lim_{t \rightarrow 0^+} [x \ln x - x] \Big|_t^1 = \lim_{t \rightarrow 0^+} [-1 - t \ln(t) + t]$$

- To find the limit of the first term we use L'Hospital's Rule:

$$= \lim_{t \rightarrow 0^+} [t \ln(t)] = \lim_{t \rightarrow 0^+} \frac{\ln(t)}{\left(\frac{1}{t}\right)} = \lim_{t \rightarrow 0^+} \frac{\left(\frac{1}{t}\right)}{\left(-\frac{1}{t^2}\right)} = \lim_{t \rightarrow 0^+} (-t) = 0$$

Hence,

$$\int_0^1 \ln(x) dx = \lim_{t \rightarrow 0^+} [-t \ln(t) - 1 + t] = -1 \quad \checkmark$$

Improper Integral of Type 2: Discontinuous Integrand

